

02 _ ROS2 개발환경 구축-2

2101080 정진용

1. ROS2를 설치하고 4~9페이지의 명령어 실행결과를 첨부하시오.

(Question)

ROS2 패키지 설치확인

□ 설치 후 다음 명령어를 실행하면 다음처럼 도움말이 출력되어야 함

```
$ source /opt/ros/foxy/setup.bash
```

```
$ ros2 --help
```

(Answer)

```
linux@JINYOUNG:~$ source /opt/ros/foxy/setup.bash
linux@JINYOUNG:~$ ros2 --help
usage: ros2 [-h] Call 'ros2 <command> -h' for more detailed usage. ...
```

```
ros2 is an extensible command-line tool for ROS 2.
```

```
optional arguments:
```

```
  -h, --help            show this help message and exit
```

```
Commands:
```

```
  action      Various action related sub-commands
  bag         Various rosbag related sub-commands
  component   Various component related sub-commands
  daemon      Various daemon related sub-commands
  doctor       Check ROS setup and other potential issues
  interface   Show information about ROS interfaces
  launch      Run a launch file
  lifecycle   Various lifecycle related sub-commands
  multicast   Various multicast related sub-commands
  node        Various node related sub-commands
  param       Various param related sub-commands
  pkg         Various package related sub-commands
  run         Run a package specific executable
  security    Various security related sub-commands
  service     Various service related sub-commands
  topic       Various topic related sub-commands
  wtf         Use 'wtf' as alias to 'doctor'
```

```
Call 'ros2 <command> -h' for more detailed usage.
```

(Question)

ROS 개발툴 설치 확인

설치 후 다음 명령어를 실행하면 다음처럼 도움말이 출력되어야 함

```
$ source /opt/ros/foxy/setup.bash
```

```
$ colcon --help
```

(Answer)

```
linux@JINYOUNG:~$ source /opt/ros/foxy/setup.bash
linux@JINYOUNG:~$ colcon --help
usage: colcon [-h] [--log-base LOG_BASE] [--log-level LOG_LEVEL]
              {build,extension-points,extensions,graph,info,list,metadata,test,te
st-result,version-check}
...

optional arguments:
  -h, --help                show this help message and exit
  --log-base LOG_BASE       The base path for all log directories (default: ./log,
                             to disable: /dev/null)
  --log-level LOG_LEVEL     Set log level for the console output, either by numeric
                             or string value (default: warning)

colcon verbs:
  build                     Build a set of packages
  extension-points          List extension points
  extensions                List extensions
  graph                    Generate a visual representation of the dependency graph
  info                     Package information
  list                     List packages, optionally in topological ordering
  metadata                 Manage metadata of packages
  test                     Test a set of packages
  test-result              Show the test results generated when testing a set of
                             packages
  version-check             Compare local package versions with PyPI

{build,extension-points,extensions,graph,info,list,metadata,test,test-result,ve
rsion-check}

call `colcon VERB -h` for specific help
```

(Question)

ROS2 패키지 예제 테스트

□ 리눅스 터미널 창을 열고 아래 명령어 실행, 종료는 Ctrl+c

```
$ source /opt/ros/foxy/setup.bash
```

```
$ ros2 run demo_nodes_cpp talker -> 'hello world: x' 문자열 전송
```

(Answer)

```
linux@JINYOUNG:~$ ros2 run demo_nodes_cpp talker
[INFO] [1757427122.799890002] [talker]: Publishing: 'Hello World: 1'
[INFO] [1757427123.799693687] [talker]: Publishing: 'Hello World: 2'
[INFO] [1757427124.799594544] [talker]: Publishing: 'Hello World: 3'
[INFO] [1757427125.799701990] [talker]: Publishing: 'Hello World: 4'
[INFO] [1757427126.799693104] [talker]: Publishing: 'Hello World: 5'
[INFO] [1757427127.799731549] [talker]: Publishing: 'Hello World: 6'
[INFO] [1757427128.799735011] [talker]: Publishing: 'Hello World: 7'
[INFO] [1757427129.799785478] [talker]: Publishing: 'Hello World: 8'
[INFO] [1757427130.799672969] [talker]: Publishing: 'Hello World: 9'
[INFO] [1757427131.799673907] [talker]: Publishing: 'Hello World: 10'
[INFO] [1757427132.799681938] [talker]: Publishing: 'Hello World: 11'
[INFO] [1757427133.799767780] [talker]: Publishing: 'Hello World: 12'
[INFO] [1757427134.799666548] [talker]: Publishing: 'Hello World: 13'
[INFO] [1757427135.799758187] [talker]: Publishing: 'Hello World: 14'
```

(Question)

ROS2 패키지 예제 테스트

- 새로운 리눅스 터미널 창을 열고 아래 명령어 실행, 종료는 Ctrl+c
\$ source /opt/ros/foxy/setup.bash
\$ ros2 run demo_nodes_py listener -> hello world: x 문자열 수신

(Answer)

```
Linux@JINYOUNG:~$ source /opt/ros/foxy/setup.bash
Linux@JINYOUNG:~$ ros2 run demo_nodes_py listener -
[INFO] [1757427141.821316837] [listener]: I heard: [Hello World: 20]
[INFO] [1757427142.801348276] [listener]: I heard: [Hello World: 21]
[INFO] [1757427143.801086849] [listener]: I heard: [Hello World: 22]
[INFO] [1757427144.800680156] [listener]: I heard: [Hello World: 23]
[INFO] [1757427145.801151889] [listener]: I heard: [Hello World: 24]
[INFO] [1757427146.801348528] [listener]: I heard: [Hello World: 25]
[INFO] [1757427147.801433367] [listener]: I heard: [Hello World: 26]
[INFO] [1757427148.801341649] [listener]: I heard: [Hello World: 27]
[INFO] [1757427149.800840831] [listener]: I heard: [Hello World: 28]
[INFO] [1757427150.801342132] [listener]: I heard: [Hello World: 29]
[INFO] [1757427151.801075662] [listener]: I heard: [Hello World: 30]
[INFO] [1757427152.801316171] [listener]: I heard: [Hello World: 31]
[INFO] [1757427153.801223406] [listener]: I heard: [Hello World: 32]
[INFO] [1757427154.801453305] [listener]: I heard: [Hello World: 33]
[INFO] [1757427155.801535624] [listener]: I heard: [Hello World: 34]
[INFO] [1757427156.801314443] [listener]: I heard: [Hello World: 35]
```

(Question)

ROS2 빌드 테스트

- ros2_ws 이라는 이름으로 워크스페이스(작업공간) 폴더를 생성한 후 빌드 명령어 colcon 실행
- 빌드 후 워크스페이스 폴더의 서브 폴더로 src 이외에도 build, install, log 폴더가 생성되는 것을 확인할 것

```
$ source /opt/ros/foxy/setup.bash
```

```
$ mkdir -p ~/ros2_ws/src
```

```
$ cd ~/ros2_ws/
```

```
$ colcon build --symlink-install
```

```
$ ls -al
```

(Answer)

```
linux@JINYOUNG:~$ mkdir -p ~/ros2_ws/src
linux@JINYOUNG:~$ cd ~/ros2_ws/
linux@JINYOUNG:~/ros2_ws$ colcon build --symlink-install

Summary: 0 packages finished [0.65s]
linux@JINYOUNG:~/ros2_ws$ ls -al
total 24
drwxr-xr-x 6 linux linux 4096 Sep  9 23:25 .
drwxr-xr-x 7 linux linux 4096 Sep  9 23:25 ..
drwxr-xr-x 2 linux linux 4096 Sep  9 23:25 build
drwxr-xr-x 2 linux linux 4096 Sep  9 23:25 install
drwxr-xr-x 3 linux linux 4096 Sep  9 23:25 log
drwxr-xr-x 2 linux linux 4096 Sep  9 23:25 src
linux@JINYOUNG:~/ros2_ws$
```


2. 11페이지 수정결과를 첨부하시오.



~/.bashrc 파일 수정

11

```
$ .bashrc
110 # sources /etc/bash.bashrc).
111 if ! shopt -oq posix; then
112   if [ -f /usr/share/bash-completion/bash_completion ]; then
113     . /usr/share/bash-completion/bash_completion
114   elif [ -f /etc/bash_completion ]; then
115     . /etc/bash_completion
116   fi
117 fi
118
119 source /opt/ros/foxy/setup.bash
120 source ~/ros2_ws/install/local_setup.bash
121
122 source /usr/share/colcon_argcomplete/hook/colcon-argcomplete.bash
123 source /usr/share/vcstool-completion/vcs.bash
124 source /usr/share/colcon_cd/function/colcon_cd.sh
125
126 export _colcon_cd_root=~/.ros2_ws
127 export ROS_DOMAIN_ID=10
128 export ROS_NAMESPACE=robot1
129 export RMW_IMPLEMENTATION=rmw_fastrtps_cpp
130 export RCUTILS_COLORIZED_OUTPUT=1
131 export RCUTILS_LOGGING_USE_STDOUT=0
132 export RCUTILS_LOGGING_BUFFERED_STREAM=1
```

(Answer)

```
$ .bashrc
89
90 # some more ls aliases
91 alias ll='ls -lF'
92 alias la='ls -A'
93 alias l='ls -CF'
94
95 # Add an "alert" alias for long running commands. Use like so:
96 # sleep 10; alert
97 alias alert='notify-send --urgency=low -i "${[ $? = 0 ]} && echo terminal || echo error)" "${history/tail -n1[sed -e '\''s/\^s*[0-9]\\\+s*//;s/[;&]]\s*alert$/'
98
99 # Alias definitions.
100 # You may want to put all your additions into a separate file like
101 # ~/.bash_aliases, instead of adding them here directly.
102 # See /usr/share/doc/bash-doc/examples in the bash-doc package.
103
104 if [ -f ~/.bash_aliases ]; then
105   . ~/.bash_aliases
106 fi
107
108 # enable programmable completion features (you don't need to enable
109 # this, if it's already enabled in /etc/bash.bashrc and /etc/profile
110 # sources /etc/bash.bashrc).
111 if ! shopt -oq posix; then
112   if [ -f /usr/share/bash-completion/bash_completion ]; then
113     . /usr/share/bash-completion/bash_completion
114   elif [ -f /etc/bash_completion ]; then
115     . /etc/bash_completion
116   fi
117 fi
118
119 source /opt/ros/foxy/setup.bash
120 source ~/ros2_ws/install/local_setup.bash
121
122 source /usr/share/colcon_argcomplete/hook/colcon-argcomplete.bash
123 source /usr/share/vcstool-completion/vcs.bash
124 source /usr/share/colcon_cd/function/colcon_cd.sh
125
126 export _colcon_cd_root=~/.ros2_ws
127 export ROS_DOMAIN_ID=0
128 export ROS_NAMESPACE=jetson0
129 export RMW_IMPLEMENTATION=rmw_fastrtps_cpp
130 export RCUTILS_COLORIZED_OUTPUT=1
131 export RCUTILS_LOGGING_USE_STDOUT=0
132 export RCUTILS_LOGGING_BUFFERED_STREAM=1
```

```
linux@JINYOUNG:~$ cd
linux@JINYOUNG:~$ source .bashrc
linux@JINYOUNG:~$ echo $ROS_DOMAIN_ID
0
linux@JINYOUNG:~$ echo $ROS_NAMESPACE
jetson0
linux@JINYOUNG:~$ █
```